EXPERIMENT 10

Applying performance optimization techniques (indexing, subforms, query tuning) to a Zoho Creator application.

Student details	Project details
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Aim

To compare record-by-record and aggregate queries, apply bounded retrieval, and evaluate indexing and subform design for the insurance application.

Execution status

Query tuning was implemented and tested through Verify_Claim_Query_Optimization. Indexing and subforms were evaluated as design considerations; no new index or subform was created. No measured latency improvement is claimed.

Application context

The project records insurance requests, validates submitted details, assigns claims for processing and presents claim status and amount summaries. Existing forms, reports, Deluge workflows and permission profiles provide the base for this experiment.

Screenshots in this report are actual captures from the project account. Explanations, tables and code listings are documentation, not simulated platform screenshots.

Method and implementation

1. Created a read-only Deluge verification function in Development. It compares a baseline loop with a criteria-based count.

2. Used count() and sum() for totals instead of computing every total in application loops.

3. Retrieved recent claims with Added_Time descending and range 0 to 4; retrieved the oldest ten open claims with ascending Added_Time and range 0 to 9.

4. Executed the function against empty Development data and then three synthetic records.

5. Changed only the synthetic Travel claim to Rejected and verified that the pending report and dashboard excluded it.

```
optimized_pending = Claim_Request_Form[
  Claim_Status != "Approved" &&
  Claim_Status != "Rejected"].count();
requested_total = Claim_Request_Form[ID != 0]
  .sum(Claim_Amount_Requested);
// Recent five: sort by Added_Time desc range from 0 to 4
// Oldest ten open: sort by Added_Time range from 0 to 9
```

Technique	Application and limitation
Query tuning - implemented	Filter in the query; aggregate counts and amounts; bound displayed lists. The baseline full scan exists only as an educational comparison.
Indexing - reviewed	Zoho documents backend/support-assisted indexing. No index was provisioned or benchmarked in this account.
Subforms - reviewed	A repeating invoice/item subform could model multiple expenses per claim. It was not added to the original basic-field form; large nested collections would require separate performance tests.

Evidence: function and tests

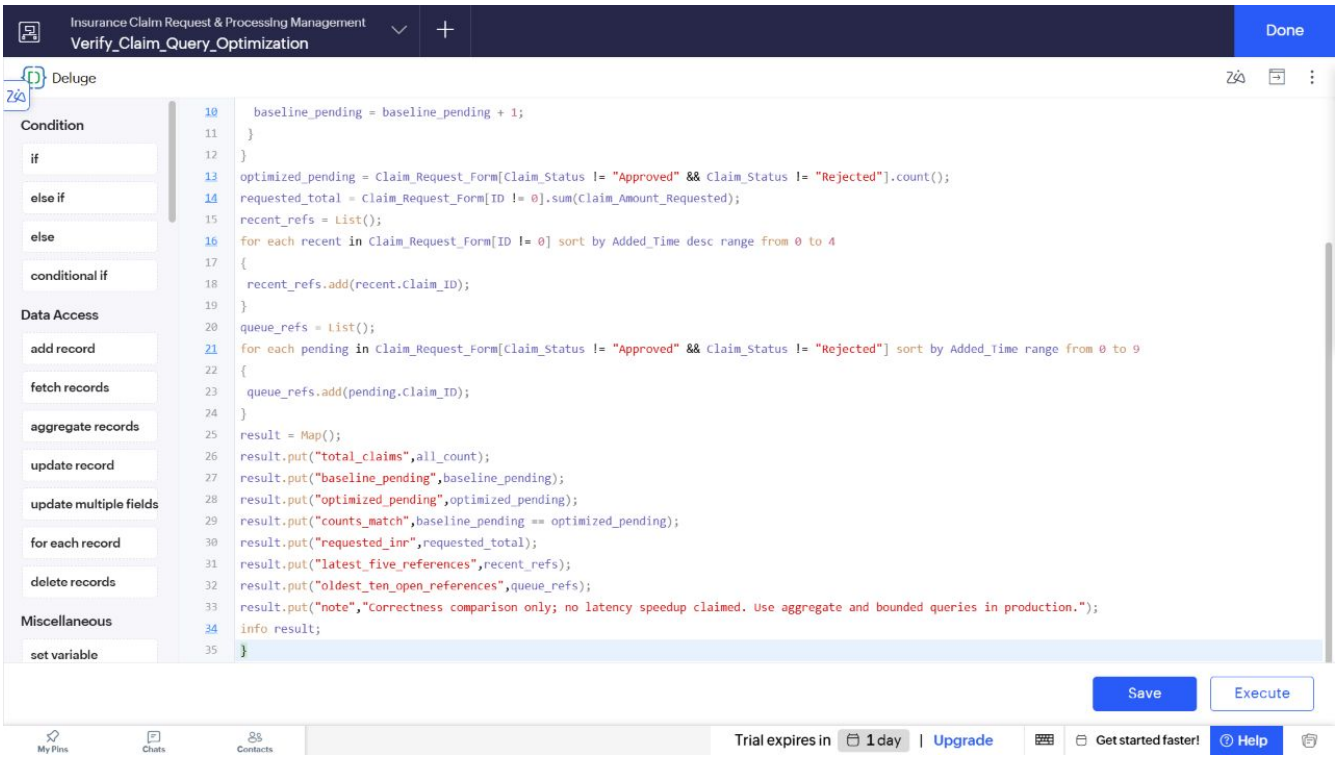


Figure 10.1 - Saved Deluge query-verification function in the actual application.

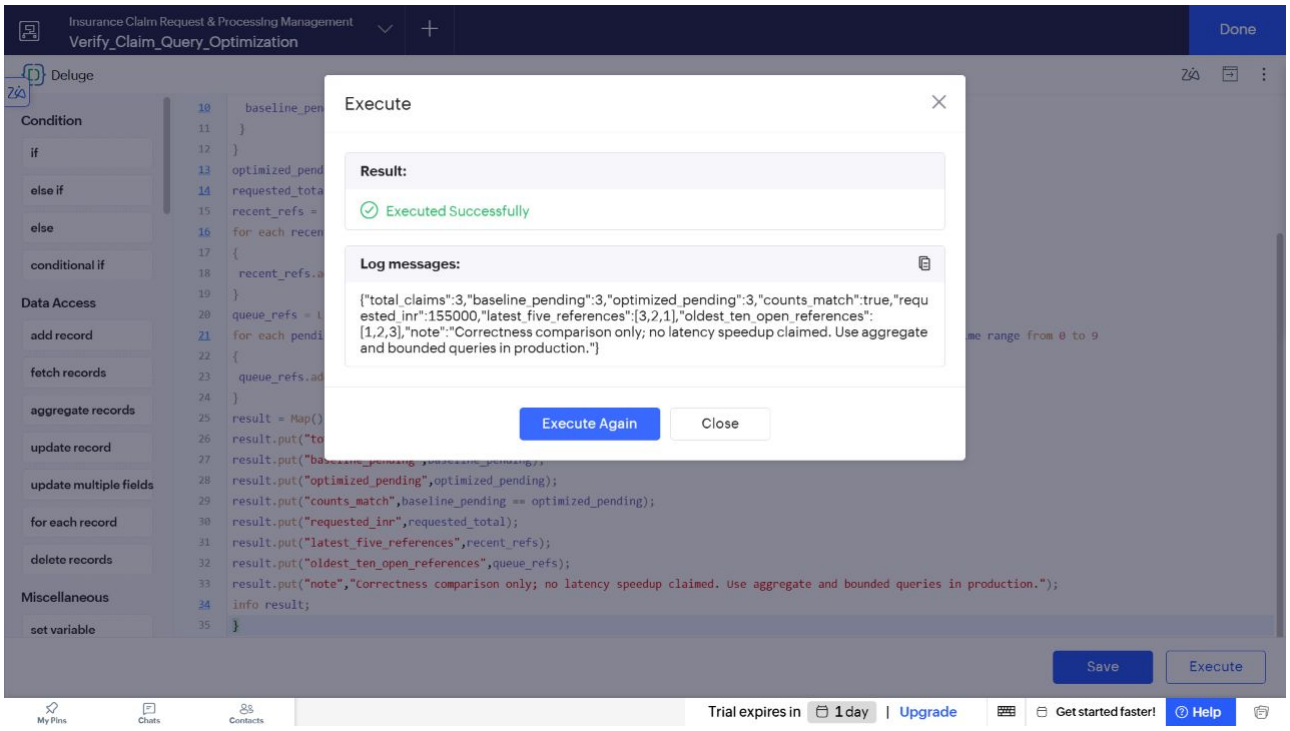


Figure 10.2 - Actual execution on three submitted demo claims: baseline and aggregate counts both equal 3.

Test results and interpretation

Test	Observed result
Empty Development dataset	0 total; 0 baseline pending; 0 optimized pending; counts_match true.
Three submitted demo claims	3 total; pending 3 in both paths; INR 155,000; references [3,2,1] recent and [1,2,3] open.
Closed-claim exclusion	Travel demo changed to Rejected; pending report shows 2 records and dashboard shows 2 pending.
Dataset isolation	Development has three synthetic records; Production has eleven existing records.
Performance claim	Functional equivalence verified. No timed load test, index comparison or throughput benchmark was performed.

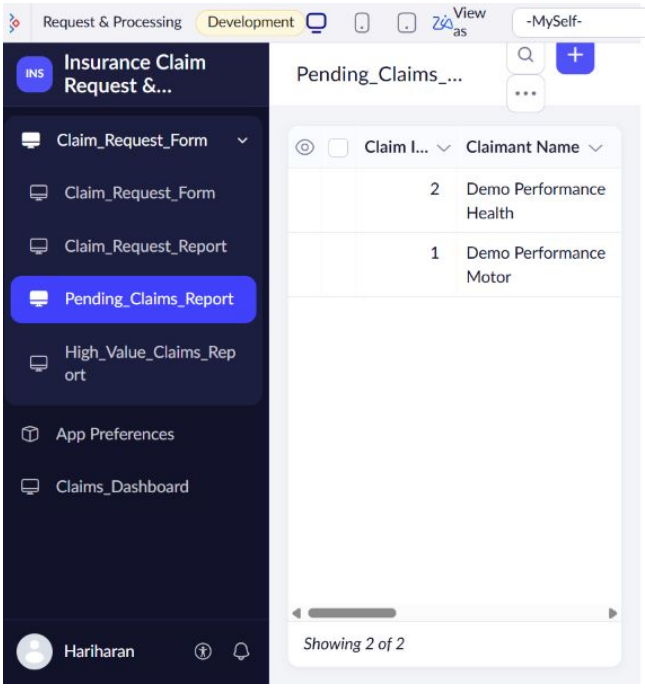


Figure 10.3 - Actual filtered report after closing the synthetic Travel claim; two open records remain.

Result

Aggregate and bounded query techniques were demonstrated. The experiment is partially implemented with respect to the full parenthetical syllabus scope: indexing and subforms remain evaluated designs rather than executed changes. A future benchmark should use representative data volumes and repeated timings under comparable conditions.

References

Zoho Creator performance guidance: https://help.zoho.com/portal/en/kb/creator/developer-guide/others/platform-performance/articles/platform-performance

Deluge fetch records: https://www.zoho.com/deluge/help/fetch-records/fetch-collection-records.html